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Experiment No: 7

Date:

Experiment Title: Design of Library Management System using UML State Chart Diagram

Aim:

To design and model a Library Management System using UML State Chart Diagram in order to represent the different states of book transactions and the dynamic behavior of the system during issue, reservation, and return processes.

Problem Description:

A Library Management System provides services such as searching books, issuing books, reserving books, and returning books. The system must verify the user details before allowing transactions. It checks the availability of books and allows issuing if available. If the book is not available, the system may allow reservation or exit the process.

Given Conditions / Constraints:

• The system should authenticate the library user before accessing services.

• The system must check book availability before issuing.

• The system should allow reservation if the book is available.

• The system must update the library database after issuing a book.

• The system should track the return date of issued books.

• The system must handle overdue conditions and apply fines if necessary.

• The process should end after successful return of the book.

Approach / Methodology:

• Identify the main states such as Login, Validation, Availability Check, Issue Book, Return Book, and Overdue.

• Define the transitions between states based on conditions like valid user, book availability, and return status.

• Model the initial state and final state of the system.

• Represent decision points such as valid/invalid user and available/not available book.

• Construct the UML State Chart Diagram to represent the dynamic behavior of the library system.

Planning and Estimation:

• Study and analyze the requirements of the Library Management System.

• Identify the different states involved in book management.

• Develop the State Chart Diagram.

• Verify the transitions between different states.

• Validate the correctness of the diagram.

Scheduling / Design Considerations:

• Ensure proper validation of library members.

• Maintain accurate book availability records.

• Prevent issuing books if they are not available.

• Clearly represent state transitions in the diagram.

• Maintain return deadlines and overdue tracking.

Technical Feasibility:

The Library Management System is technically feasible as it follows standard software design principles and UML modeling techniques.

Operational Feasibility:

The system is easy to operate for both library staff and users. It simplifies book tracking, issuing, and returning processes.

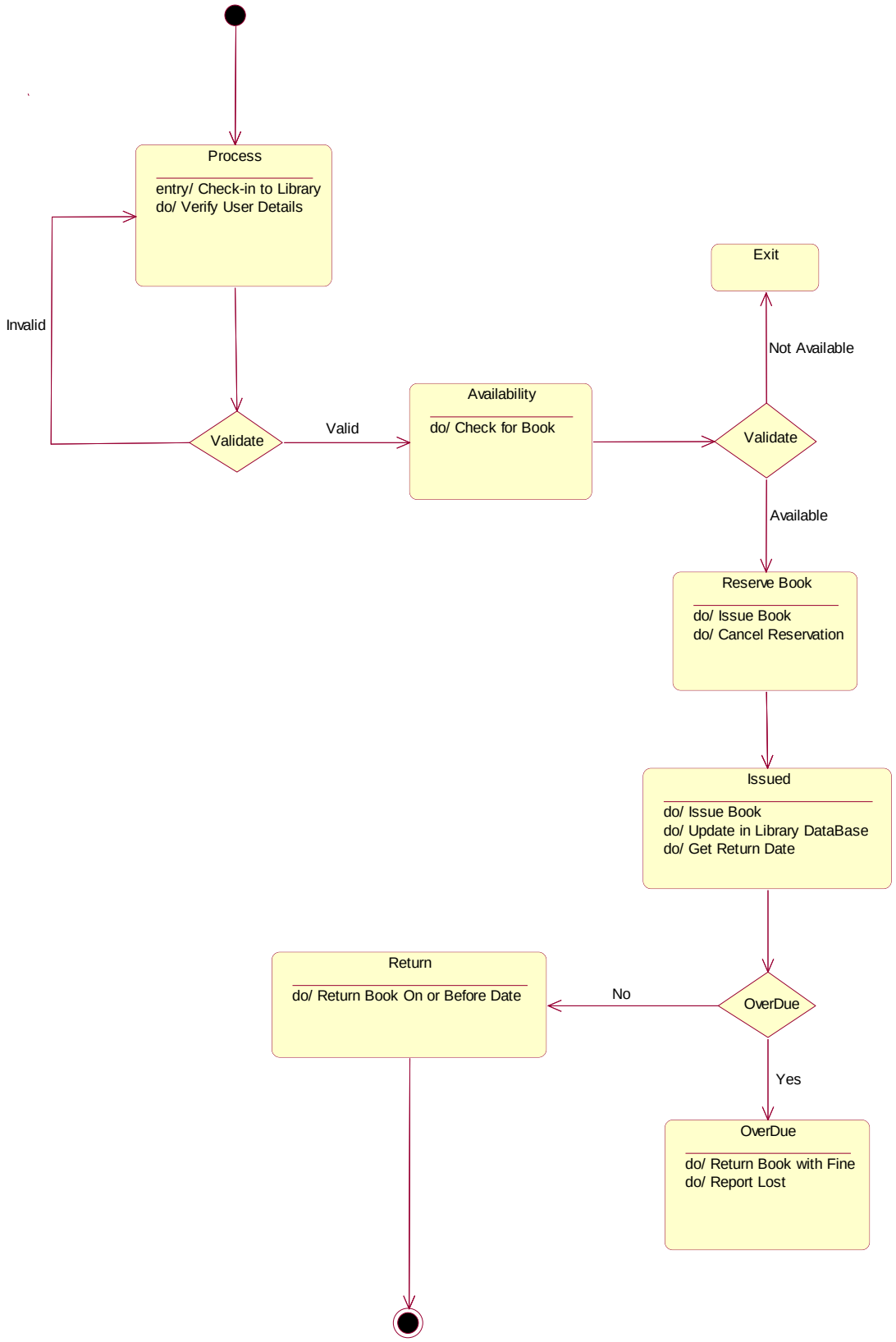
Economic Feasibility:

The system reduces manual record keeping and improves efficiency, making it cost-effective for library management.

Result:

The Library Management System was successfully modeled using a UML State Chart Diagram. The diagram clearly shows the different states such as validation, availability checking, issuing, returning, and overdue handling. It also shows the transitions between these states based on conditions.

Conclusion:

The experiment demonstrates the modeling of a Library Management System using a UML State Chart Diagram. The diagram explains the different states involved in the book issue and return process. It helps in understanding the dynamic behavior of the system. This model helps developers design an efficient and reliable library management system.

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